UNIVERSITY INITIATIVES COMPETITION

Category: Student-Led Academic Initiatives

Youth Computing Research Journal

A Structured Pathway for University Students
from First Principles to Published Research

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Confidential Initiative Proposal

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1. Executive Summary

Initiative at a Glance

Initiative Name:	Youth Computing Research Journal (YCRJ)
Target Audience:	Undergraduate students with no prior research experience
Core Mission:	A complete, structured pathway from zero knowledge to peer-reviewed publication
Structure:	Three progressive phases — Foundations, Research & Writing, Publication
Primary Output:	A student-authored, institutionally indexed journal of computing research

The **Youth Computing Research Journal (YCRJ)** is a student-founded academic initiative designed to address a clear and persistent gap in undergraduate education: the absence of a structured, end-to-end pathway that equips junior researchers with the skills, tools, and guided experience necessary to produce publishable scientific work in computing.

Most undergraduate students possess curiosity and technical ability, yet lack exposure to the norms of academic writing, the mechanics of scientific paper structure, the professional tools used in research documentation, and the practical process of navigating peer review and publication. As a result, meaningful undergraduate research contributions remain rare — not due to lack of talent, but due to lack of guidance.

YCRJ directly resolves this by offering a three-phase program: a structured foundational curriculum, a supervised research and writing sprint, and a publication pipeline that results in an indexed, creditable journal entry. Every student who completes the program will have gone through a complete research lifecycle — from learning what a paper is, to having their name on one.

2. Problem Statement

The gap between undergraduate education and active research participation is wide, particularly in computing disciplines. Students encounter the following systemic barriers:

- **No formal research writing curriculum:** Undergraduate programs typically teach programming and theory but rarely include structured training in how to write, structure, and submit a scientific paper.
- **Unfamiliarity with academic tools:** Tools such as LaTeX (for typesetting), Matplotlib (for data visualization), and reference management software are either never introduced or left entirely to self-study.
- **No access to mentorship pipelines:** Junior students rarely have organic access to senior researchers or graduate students who can guide their first research attempt.
- **Academic integrity gaps:** Concepts such as plagiarism, proper citation, and research ethics are frequently underdeveloped at the undergraduate level.

- **Absence of a publishing venue:** Even when students produce quality work, there is no accessible, credible venue specifically designed for undergraduate computing research.
- **Missed downstream opportunities:** Without a publication or documented research experience, students are disadvantaged when applying for graduate programs, research internships, or academic competitions.

YCRJ is built specifically to dismantle each of these barriers through a deliberate, well-resourced, and community-driven program.

3. Initiative Objectives

Primary Objective: To establish the first university-level youth computing research journal that operates as both a training program and a publication platform — enabling undergraduates to learn, write, and publish original computing research.

Supporting Objectives:

1. Deliver a modular, accredited foundational curriculum covering all skills required for scientific writing in computing.
2. Produce a structured 5-week supervised research and writing sprint resulting in complete paper submissions.
3. Establish a transparent, rubric-based peer review and ranking system aligned with international academic standards.
4. Publish the highest-quality submissions in YCRJ's indexed, publicly accessible journal.
5. Build a sustainable student organization with institutional recognition that grows each academic cycle.
6. Provide students with a creditable, citable publication that strengthens their graduate school, internship, and competition applications.
7. Create a replicable model that can expand to other departments and potentially other universities.

4. Program Structure Overview

YCRJ is organized into three sequential, interdependent phases. Each phase has defined learning outcomes, deliverables, timelines, and assessment criteria. Participation in Phase 2 is contingent on completion of Phase 1. Publication in Phase 3 is conditional on meeting the criteria established in Phase 2.

*Total Program Duration: Approximately **14–17 Weeks** per Annual Cohort*

★ Capacity Guideline

To maintain quality of instruction and feedback, each cohort is limited to **a maximum of 15 participants per coordinator**. At this scale, reviewing weekly deliverables requires approximately 5–8 hours per week. Exceeding this number significantly degrades feedback quality and student outcomes.

5. Phase 1: Foundational Program

5.1 Overview and Purpose

Phase 1 is the backbone of YCRJ. It is a structured, modular educational program designed to take a student with no prior research or academic writing experience and equip them with every skill they need to write, format, and contextualize a computing research paper. Upon completion, participants will possess both the conceptual understanding and the practical tool-based competencies required to enter Phase 2 with confidence.

Phase 1 consists of **eight modules**, each targeting a distinct and essential competency. Modules are delivered through a combination of synchronous sessions (workshops and labs), asynchronous resources (recorded lectures, reading materials, and exercises), and formative assessments.

5.2 Duration

Phase 1 runs for **6 to 8 weeks**, with approximately 2 contact hours per week plus self-directed study time. This schedule is designed to be compatible with students' regular academic workloads.

5.3 Diagnostic Entry Assessment

In the **first session**, all applicants complete a short 5-question diagnostic quiz covering basic concepts (e.g., the difference between a research paper and a report; what a journal impact factor is). Students who score very poorly are advised to complete a preliminary academic reading course before joining, ensuring that the cohort starts from a shared baseline and protecting program completion rates.

5.4 Module Curriculum

Module 1 — What Is a Scientific Paper? **(1 week)**

Objective: To give students a clear, concrete understanding of what a scientific paper is, why it exists, and how it is read and used by the research community.

Topics: types of papers (original research, surveys, short papers, technical reports); the anatomy of a computing paper; how to read a paper efficiently (the three-pass method); how reviewers evaluate papers.

Deliverable: Students annotate and summarize one assigned computing paper using a structured template, identifying each section and its function.

Module 2 — Academic and Scientific English Writing (1 week)

Objective: To develop students' proficiency in the precise, formal, and objective register of academic scientific writing.

Topics: clarity, conciseness, and precision; common errors (vague language, first-person overuse); hedging language; transitional structure; writing for an international technical audience.

Deliverable: A written exercise where students improve three provided weak paragraphs from anonymized student drafts.

Module 3 — Paper Structure and Section Writing (1.5 weeks)

Objective: To teach students how to write each section of a paper — not just what goes where, but why and how.

Topics: Abstract (structured, 150–250 words); Introduction (funnel structure — broad context to specific gap to contribution); Related Work (thematic grouping, critical citation); Methodology (reproducibility); Results (objective presentation); Discussion (interpretation, limitations, implications); Conclusion (no new content); References.

Deliverable: Students write a structured outline for a hypothetical paper on an assigned computing topic, completing a section-by-section template.

Module 4 — LaTeX for Research Document Preparation (1.5 weeks)

Objective: To introduce students to LaTeX — the industry-standard typesetting system used in computing, mathematics, and engineering research — and enable them to format a complete paper.

LaTeX is the required or strongly preferred format for submissions to virtually all major computing conferences and journals (IEEE, ACM, Springer, Elsevier). It produces typographically superior output, handles mathematical notation elegantly, and automates referencing, numbering, and layout.

Topics: Overleaf setup; document structure; figures, tables, equations; BibTeX bibliography management; IEEE and ACM official templates; compiling and resolving errors.

Deliverable: Students reproduce a provided 2-page sample paper in LaTeX using the IEEE template, including a figure, a table, and a bibliography.

Module 5 — Data Visualization with Matplotlib (1 week)

Objective: To teach students how to create professional, publication-quality figures using Matplotlib.

Matplotlib is the dominant Python library for scientific data visualization. Proper visualization communicates results more accurately and convincingly than text or tables alone.

Topics: line plots, bar charts, scatter plots, histograms; customizing axes, legends, color schemes; principles of scientific visualization; exporting at 300+ DPI in PDF/SVG format for LaTeX compatibility.

Deliverable: Students receive a provided dataset and produce three distinct, properly labeled publication-quality figures, embedded in their LaTeX document from Module 4.

Module 6 — Citations, References, and Academic Integrity **(0.5 week)**

Objective: To establish a thorough and practical understanding of citation practice, plagiarism, and research ethics.

Topics: types of plagiarism (verbatim, paraphrasing, idea theft); when to cite; citation styles (IEEE, ACM, APA); reference management with Zotero, Mendeley, and BibTeX; finding credible sources (Google Scholar, IEEE Xplore, ACM DL, arXiv); Turnitin similarity checking.

Deliverable: A short quiz on citation scenarios plus a BibTeX reference list constructed for a provided set of sources.

Module 7 — Research Pathways and Opportunities **(0.5 week)**

Objective: To contextualize academic writing skills within the broader landscape of opportunities available to junior researchers.

Topics: academic publishing and researcher profiles; graduate studies preparation; open-source contribution; academic competitions (IEEE, ACM, ICPC); thesis writing; technical project documentation.

Format: Panel discussion with invited alumni, graduate students, or faculty, supplemented by a written resource guide.

Module 8 — Formative Assessment and Phase 1 Completion **(0.5 week)**

Objective: To consolidate learning, identify gaps, and certify students as ready to enter Phase 2.

Activities: comprehensive review session; completion check of all module deliverables; individual or group feedback session with program coordinators.

Progression gate: Students must complete all module deliverables to a satisfactory standard to proceed to Phase 2. Students who require additional time may be offered one revision opportunity per deliverable.

5.5 Phase 1 Curriculum Summary

Table 1: Phase 1 Module Overview

#	Module	Key Competency	Duration
1	What Is a Scientific Paper?	Paper anatomy, reading strategies	1 week
2	Academic & Scientific English	Precision, clarity, register	1 week
3	Paper Structure & Section Writing	IMRaD writing, section-by-section	1.5 weeks
4	LaTeX Document Preparation	Typesetting, IEEE/ACM templates	1.5 weeks
5	Data Visualization (Matplotlib)	Publication-quality figures	1 week
6	Citations & Academic Integrity	Citation styles, plagiarism, Zotero	0.5 week
7	Research Pathways & Opportunities	Grad school, open-source, competitions	0.5 week
8	Formative Assessment	Completion check, progression gate	0.5 week
	Total Duration		**7.5 weeks**

5.6 Coordinator Weekly Responsibilities (Phase 1)

The following table provides a precise, week-by-week operational guide for the YCRJ curriculum coordinator throughout Phase 1. Sessions are recommended to be held on a fixed day and time each week (e.g., every Tuesday, 3–5 PM) to establish a consistent rhythm.

Table 2: Coordinator Tasks During Phase 1

Week	Coordinator Actions
1	Administer diagnostic entry quiz (5 questions); deliver Module 1 session (90 min): paper anatomy, types of papers, three-pass reading method; distribute LaTeX template and reading list; assign Week 1 deliverable (paper annotation).
2	Review Week 1 deliverables (5 min per student); deliver Module 2 session: academic English, precision, hedging language; hold 30-min group review of best and weakest submissions (anonymized); assign Week 2 deliverable (paragraph rewriting exercise).
3	Review Week 2 deliverables; deliver Module 3 session (Part 1): Abstract and Introduction writing with funnel structure; show real computing paper examples section by section; assign structured outline deliverable.

Week	Coordinator Actions
4	Continue Module 3 (Part 2): Related Work, Methodology, Results, Discussion, Conclusion, References; review first 5 student outlines live in session; send written feedback on remaining outlines by end of week.
5	Deliver Module 4 session (Part 1): Overleaf setup, document structure, sections, figures, tables; conduct hands-on lab — every student compiles their first LaTeX document live; assign IEEE template reproduction deliverable.
6	Continue Module 4 (Part 2): BibTeX, bibliography styles, common errors; review LaTeX deliverables; deliver Module 5 session: Matplotlib workflow, plot types, publication standards; assign Matplotlib figures deliverable (3 figures from provided dataset).
7	Deliver Module 6 session: plagiarism types, citation styles, Zotero/BibTeX workflow, Turnitin; assign citation quiz and BibTeX reference list; deliver Module 7 as panel discussion with invited guests or recorded alumni panel.
8	Module 8 — Formative Assessment: comprehensive review session; completion check of all deliverables; individual or group feedback session; issue Phase 1 Completion Certificates to qualifying students; announce Phase 2 group assignments.

★ Continuous Coordinator Responsibilities (All Weeks)

Throughout Phase 1, the coordinator must also:

- Respond to student inquiries via the group channel **within 24 hours**.
- Record attendance each session and flag students approaching the 3-absence limit.
- Maintain a backup copy of every submitted deliverable in an organized cloud folder.
- Send a reminder **24 hours before each session** with the agenda and any pre-reading.
- Return graded deliverables within **3 working days** of the submission deadline.

5.7 Phase 1 Student Deliverables Schedule

Table 3: Phase 1 Weekly Deliverables

Week	Deliverable	Content	Format
1	Paper Annotation	Annotated summary of one assigned computing paper; identify each section and its function	PDF
2	Writing Exercise	Three rewritten paragraphs from provided weak drafts; brief justification of changes	DOCX
3–4	Structured Outline	Complete section-by-section outline for a hypothetical computing paper using the provided template	DOCX
5–6	LaTeX Reproduction	2-page paper in IEEE template with one figure, one table, BibTeX bibliography, and Matplotlib figures embedded	PDF
7	Citation Quiz & BibTeX	Short quiz on citation scenarios; BibTeX reference list for a provided set of 5 sources	PDF
8	Completion Package	All prior deliverables revised to satisfactory standard; submitted as a single organized folder	ZIP

6. Phase 2: Research and Writing Sprint

6.1 Overview and Purpose

Phase 2 is where foundational knowledge becomes applied output. Students are organized into small research groups, each paired with a Senior Researcher Mentor, and given five weeks to conceive, develop, write, and submit an original computing research paper. This phase replicates the experience of an actual research sprint — with real constraints, real collaboration, and a real submission deadline.

6.2 Group Formation

- Groups consist of **3 to 4 junior researchers** (Phase 1 graduates).
- Each group is assigned one **Senior Researcher Mentor** — a graduate student, advanced undergraduate with research experience, or faculty-affiliated researcher.
- Groups are formed by the YCRJ organizing committee, with consideration given to complementary skills and scheduling compatibility.
- Each group must designate a **Group Lead** responsible for coordinating meetings and serving as the primary contact with their mentor.

6.3 Research Question Selection

Each group selects one research question from within the field of computing. Topics may span any subfield: artificial intelligence, computer vision, NLP, cybersecurity, systems and networks, software engineering, HCI, data science, algorithms, and more.

The research question must be: (1) original, (2) feasible within the 5-week timeline, and (3) scoped appropriately for a short paper (4–8 pages in IEEE format). Groups submit their proposed question and a brief justification for approval within the **first three days** of Phase 2.

6.4 The Five-Week Timeline

Table 4: Phase 2 Weekly Milestones

Week	Activities and Milestones
1	Research question submission and approval; literature review; defining methodology
2	Data collection, experimentation, or analysis; Related Work section draft
3	Results generation; Methodology and Results sections drafted; **first mentor review**
4	Introduction, Discussion, and Conclusion drafted; full paper draft submitted for mentor feedback
5	Revisions; LaTeX formatting; abstract finalization; **final submission**

6.5 Coordinator Weekly Responsibilities (Phase 2)

The following table provides coordinators with a precise, week-by-week operational guide throughout the sprint:

Table 5: Coordinator Tasks During Phase 2

Week	Coordinator Actions
1	Confirm group formation and mentor assignments; verify research question submissions; send approval or revision requests within 48 hours
2	Check in with all mentors; flag any groups that have not begun literature review; send reminder of Week 3 mentor review requirement
3	Receive and log first mentor review reports; identify any groups at risk of scope creep or timeline slippage; offer optional group check-in sessions
4	Receive full paper drafts; distribute to mentors for structured feedback; track submission completeness
5	Receive final submissions; run Turnitin similarity checks; confirm all submission requirements are met (PDF, similarity report, contribution statement, declaration form)

6.6 Mentor Responsibilities

Senior Researcher Mentors are expected to:

- Meet with their assigned group **at least once per week** for a structured progress check.
- Review section drafts and provide written feedback **within 48 hours** of receipt.

- Ensure the group's work remains original and properly attributed.
- Flag any academic integrity concerns immediately to the program committee.
- Provide a brief written assessment of the group's collaboration and effort at the end of Phase 2.

Mentors are not co-authors. Their role is advisory. Authorship belongs exclusively to the junior researcher group members.

6.7 Submission Requirements

At the end of Week 5, each group submits:

1. A complete paper in **IEEE double-column format**, compiled in LaTeX, as a PDF — between 4 and 8 pages (excluding references).
2. A **similarity report** generated via Turnitin or equivalent, demonstrating no more than 15% similarity.
3. A **contribution statement** declaring the specific contribution of each group member.
4. A completed **author declaration form** affirming originality and agreement with YCRJ publication terms.

7. Phase 3: Review, Ranking, and Publication

7.1 Overview and Purpose

Phase 3 closes the research lifecycle. Submitted papers undergo a structured review process modeled on international academic publishing standards. Papers are evaluated against a published rubric, ranked, and the highest-quality submissions are selected for publication in YCRJ's indexed journal. This phase gives students an authentic experience of peer review.

7.2 The YCRJ Review Process

Each submitted paper is independently evaluated by two reviewers: one **student reviewer** (a Phase 1 graduate not affiliated with the submitting group) and one **senior reviewer** (a mentor or faculty-affiliated reviewer). Reviews are conducted **blind** — reviewers do not know the identity of the authors.

Scores from both reviewers are averaged to produce the paper's final score. In the case of a significant score discrepancy (more than 20 points on a 100-point scale), a third senior reviewer is assigned as a tiebreaker.

7.3 YCRJ Evaluation Rubric

Table 6: YCRJ Peer Review Rubric (100 Points Total)

Criterion	Max Points	What Is Assessed
Research Question Clarity	10	Is the question clearly stated, specific, and original?
Literature Review Quality	10	Is related work adequately covered and critically engaged with?
Methodology Rigor	15	Is the approach described clearly and reproducibly?
Results Quality	15	Are results presented accurately, with appropriate visualizations?
Discussion and Analysis	15	Does the paper interpret results insightfully and acknowledge limitations?
Writing and Language	10	Is the paper written clearly and in appropriate academic English?
LaTeX Formatting	10	Does the paper conform to IEEE format with correct figures and citations?
Originality and Contribution	15	Does the paper make a clear, original contribution?
Total	**100**	

7.4 Student Reviewer Form

The following one-page form is completed by student reviewers for each assigned paper:

Reviewer Name: _____ **Date:** _____
Paper Code: _____

Section 1: Quick Checklist *(tick where applicable)*

- ☐ Title is clear and contains key variables
- ☐ Abstract contains (background – aim – methods – results – conclusion)
- ☐ Introduction clearly identifies the research gap
- ☐ Methodology is described as reproducible steps
- ☐ Results are presented with clear figures/tables produced in Matplotlib or equivalent
- ☐ Discussion compares results with previous studies (at least 3 comparisons)
- ☐ In-text citations exist and reference list is complete
- ☐ LaTeX formatting conforms to IEEE template

Section 2: Two Strongest Points

1. _____
2. _____

Section 3: Two Points Requiring Improvement (with reason)
1. _____
Reason: _____
2. _____
Reason: _____

Section 4: One Suggestion if You Were the Author

_____

Reviewer Signature: _____

7.5 Publication Eligibility Criteria

A paper is eligible for publication in YCRJ if it meets *all* of the following conditions:

1. Achieves a final averaged rubric score of **70 or above** out of 100.
2. Meets the **15% or below** similarity threshold on the Turnitin report.
3. Passes the formatting check for **IEEE LaTeX compliance**.
4. Has no unresolved academic integrity concerns flagged during review.
5. All authors have signed and submitted the **author declaration form**.

Of all papers meeting these criteria, the **top 3 papers** by final rubric score will be selected for publication in the inaugural issue of YCRJ. Conditional acceptance may be granted to papers scoring between 65 and 69, pending minor revisions within a 5-day window.

7.6 Author Feedback

Every submitting group receives a complete, written review report regardless of publication outcome. This includes scores per criterion, written comments from both reviewers, and suggestions for improvement. The review process is a learning experience even for groups whose papers are not selected for publication.

8. The YCRJ Journal and Publication Platform

8.1 Journal Identity and Scope

The Youth Computing Research Journal (YCRJ) will be established as a peer-reviewed, open-access academic journal publishing original computing research authored by undergraduate students. The journal's scope encompasses all subfields of computing.

8.2 Website and Digital Infrastructure

YCRJ will maintain a dedicated website serving as the journal's public presence and operational hub, including:

- A public journal archive displaying all published papers (HTML and PDF) with full metadata (title, authors, abstract, keywords, DOI)
- An author submission portal for Phase 2 paper submissions

- A program information hub covering Phase 1 curriculum resources and registration
- An editorial board page listing senior reviewers, mentors, and organizing committee members
- A guidelines and standards page publishing the evaluation rubric, submission requirements, and publication policy

The website will be built using a modern, maintainable web stack and hosted on a reliable platform (e.g., GitHub Pages, Vercel, or institutional web hosting).

8.3 DOI Assignment and Indexing Strategy

Table 7: Indexing Roadmap

Phase	Milestone	Description
A	DOI Assignment	Via Zenodo (CERN), Figshare, or institutional repository
B	Google Scholar	Scholar-compliant metadata and HTML article pages
C	ISSN Registration	From the relevant national ISSN center
D	Database Indexing	ROAD, DOAJ, Dimensions, Semantic Scholar; long-term: Scopus, WoS

9. Content Accreditation and Curriculum Credentialing

9.1 Credentials Issued to Participants

Students who successfully complete Phase 1 will receive a **YCRJ Phase 1 Completion Certificate**, issued digitally and verifiable via the YCRJ website or a digital credential platform (e.g., Credly or Badgr). The certificate will be:

- Signed by the YCRJ organizing committee leads and, upon securing institutional endorsement, countersigned by a faculty sponsor or departmental head.
- Descriptive of competencies, listing the specific skills certified (academic writing, LaTeX typesetting, Matplotlib visualization, citation management, research methodology).

Students whose papers are published in YCRJ will additionally receive a **Publication Acknowledgment Letter** suitable for graduate school applications, citing the paper by title, authors, issue, and DOI.

9.2 Institutional Accreditation Pathway

YCRJ seeks formal recognition from the university through:

- **Student organization registration** with the university's student or academic affairs office.
- **Faculty sponsorship** from one or more faculty members in the computing department.

- **Co-curricular credit recognition** on the official university transcript (subject to university policy).
- **Departmental endorsement** from the computing or engineering faculty.

10. Organizational Structure and Team

Table 8: YCRJ Organizational Roles

Role	Responsibilities
Co-Founders / Directors	Overall initiative leadership, institutional liaison, strategic direction, external partnerships
Curriculum Coordinator	Designing and maintaining Phase 1 module content, scheduling sessions, tracking student progress
Operations Coordinator	Managing submissions, communications, logistics, website updates, document processing
Editorial Board	Phase 3 review coordination, reviewer assignment, rubric enforcement, final publication decisions
Senior Researcher Mentors	Phase 2 group guidance, draft review, academic integrity monitoring, mentor assessment
Faculty Advisor	Institutional representation, quality oversight, accreditation support, escalation point
Web and Media Team	Website development, digital publishing, social media, outreach

11. Timeline and Rollout Plan

Table 9: YCRJ Launch Timeline — Cohort 1

Period	Milestones
Month 1	Secure institutional recognition; recruit faculty advisor; finalize curriculum materials; launch website
Month 2	Open student registration for Cohort 1; begin Phase 1 delivery; recruit Senior Researcher Mentors
Months 3–4	Complete Phase 1 (all 8 modules); issue Phase 1 Completion Certificates; form Phase 2 groups
Months 4–5	Phase 2 Research Sprint (5 weeks); mentor check-ins; paper submissions received
Month 6	Phase 3 review and ranking; revision window; publication selection; DOI assignment
Months 6–7	Launch inaugural issue on the YCRJ website; ISSN application; Google Scholar indexing
Month 7+	Retrospective and program improvement; open Cohort 2 registration; pursue database indexing

12. Practical Guidance: Avoiding Common Failure Modes

The following guidance is derived from operational experience running structured academic training programs. It is intended for the YCRJ organizing team and coordinators.

1. **Start with a diagnostic entry test.** In the first session, administer a 5-question quiz (e.g., what is the difference between research and a report? what is a journal's impact factor?). Students who answer completely incorrectly should be advised to take a preliminary course first.

2. **Strictly enforce the 15-participant cap.** Weekly deliverable review for 15 students takes 5–8 hours per coordinator. Exceeding this number degrades feedback quality and is a leading cause of program failure.

3. **Use simple, proven tracking tools.** An Excel sheet for attendance and grades, a messaging group for urgent announcements, and a shared cloud folder for assignment submission are sufficient. Do not introduce complex systems that create overhead.

4. **Enforce the attendance policy.** Do not allow more than **3 absences**. Absences lead to backlog and loss of motivation. Students who miss sessions must watch available recordings and submit assignments on time.

5. **Provide immediate feedback.** Assignments submitted on Sunday should be graded and returned by Wednesday. Any delay in feedback kills student motivation and breaks the weekly rhythm.

6. **Offer an optional weekly support session.** A one-hour optional session via video call, open to students who are struggling, prevents the accumulation of unresolved questions that would otherwise cause dropout.

7. **Require a commitment pledge on Day 1.** Have all participants sign a brief commitment pledge covering attendance and on-time submission. This significantly reduces mid-program dropout.

8. **Use examples from students' own computing subfields.** If a student is interested in security, use a security paper as an example. Generic examples reduce engagement. Domain-relevant examples make the skills feel immediately applicable.

9. **Prepare fallback options for every technical dependency.** If LaTeX compilation fails, teach the student to use Overleaf. If Matplotlib causes environment issues, allow an alternative (e.g., Excel charts or any plotting tool), and focus on the visualization principles. Do not allow technical failures to block content progress.

10. **Celebrate visible milestones.** After completing the literature review, send a group acknowledgment message. After submitting the first full draft, issue a *"First Draft Completed"* micro-certificate. These small recognitions significantly improve cohort retention and morale.

11. **Set honest expectations about publication outcomes.** Do not promise publication in a high-impact or Scopus-indexed journal. Communicate clearly: *"We will help you produce the strongest paper possible and submit it to YCRJ. It may be accepted as-is, require revisions, or not meet the publication threshold — and that is a normal part of the research process."*

12. **Document everything for Cohort 2.** Keep records of all coordinator decisions, student questions, module feedback, and review outcomes. The second cohort benefits enormously from the lessons of the first.

13. Applied Example: A YCRJ-Ready Paper

The following is a hypothetical student paper that illustrates the expected output of a Phase 2 group. It demonstrates the paper structure, writing register, and level of rigor YCRJ targets.

13.1 Paper Section Structure Reference

Before reviewing the example, the table below defines the mandatory structure for all YCRJ submissions. Coordinators should distribute this table to all Phase 2 groups at the start of the sprint.

Table 10: YCRJ Mandatory Paper Structure

Section	% of Paper	Word Count	Mandatory Elements
Title	—	max 15 words	Contains: main method/approach, dependent variable, population or context
Abstract	10%	150–250 words	Structured: Background (2 sentences), Aim (1 sentence), Methods (3 sentences), Results (2 sentences), Conclusion (1 sentence)
Introduction	15%	350–500 words	Three paragraphs: (1) general background, (2) related work and gap, (3) research aim and questions
Related Work	10%	250–350 words	Thematic grouping of prior work; critical positioning; every claim cited
Methodology	15%	300–450 words	Subheadings: design, dataset/participants, tools, procedures, analysis, ethical considerations
Results	20%	300–500 words	At least two figures or tables (Matplotlib); descriptive text only — no interpretation
Discussion	25%	500–700 words	Five paragraphs: (1) summary of main findings, (2) comparison with prior work, (3) interpretation of unexpected findings, (4) limitations, (5) recommendations
Conclusion	5%	100–150 words	One paragraph: what was found, why it matters, one future direction
References	—	15–25 sources	80% from last five years; formatted in IEEE style via BibTeX

13.2 Hypothetical Student Paper

Title: *Comparing Lightweight Intrusion Detection Approaches for Resource-Constrained IoT Gateways*

Abstract:
Background: IoT deployments in smart environments generate continuous network traffic that must be monitored for intrusions without imposing prohibitive computational overhead on gateway hardware. *Aim:* This study evaluates three lightweight anomaly detection algorithms — Isolation Forest, One-Class SVM, and a threshold-based statistical detector — on a public IoT traffic dataset. *Methods:* We implemented and benchmarked each algorithm on the N-BaIoT dataset using a Raspberry Pi 4 as a representative gateway device, measuring detection accuracy, false positive rate, CPU usage, and memory footprint. *Results:* Isolation Forest achieved the highest F1-score (0.91) with acceptable CPU overhead (28% average), outperforming One-Class SVM (F1: 0.86, CPU: 34%) and the statistical detector (F1: 0.79, CPU: 12%). *Conclusion:* Isolation Forest represents a practical balance

between detection performance and resource consumption for constrained IoT gateways; deployment feasibility on lower-spec hardware warrants further investigation.

Introduction (first paragraph):
The proliferation of Internet of Things devices in smart home and industrial automation contexts has expanded the attack surface of local networks considerably. Unlike traditional endpoints, IoT gateways must enforce security policies under strict constraints of CPU, memory, and power budget. Conventional intrusion detection systems designed for server-class hardware are therefore impractical in this context, and the research community has responded with a growing body of lightweight detection approaches. However, most comparative evaluations are conducted in simulation or on desktop hardware, leaving open the question of which methods perform acceptably on actual constrained gateway hardware running live traffic. This study addresses that gap directly.

14. Impact and Value Proposition

14.1 For Participating Students

- A verifiable, creditable research publication with a DOI — often a first in an undergraduate student's portfolio.
- Mastery of professional research tools (LaTeX, Matplotlib, Zotero) used by the global research community.
- Directly transferable skills for graduate school applications, thesis writing, and internship portfolios.
- Experience of a real research lifecycle: question formulation, methodology design, writing, review, and publication.
- Membership in a peer research community, with ongoing mentorship access.

14.2 For the University

- A student-led initiative that produces measurable academic output (indexed publications).
- Enhanced undergraduate research activity, benefiting department and university research profiles.
- A pipeline that prepares students for graduate research programs, increasing postgraduate enrollment quality.
- A replicable model that can scale to other departments and serve as a university flagship initiative.
- A positive signal for accreditation bodies that the institution invests in undergraduate research development.

14.3 For the Broader Computing Community

- A new, open-access venue for emerging junior research in computing.
- A talent pipeline of well-prepared junior researchers entering graduate programs and industry.
- A model for how universities can systematically lower the barrier to undergraduate research publishing.

15. Required Documents Checklist

Before the first session, coordinators must ensure the following documents exist, are backed up to the cloud, and are ready for distribution:

Table 11: Pre-Launch Document Checklist

#	Document	Description	Format
1	LaTeX Paper Template	IEEE double-column, preformatted, with figure/table placeholders	.tex
2	Module 1 Slides	Paper anatomy, reading strategies, 20–25 slides	PPTX
3	Module 2 Slides	Academic writing principles and exercises	PPTX
4	Module 3 Slides	Section-by-section writing with real examples	PPTX
5	Module 4 Slides	LaTeX/Overleaf walkthrough with screenshots	PPTX
6	Module 5 Slides	Matplotlib workflow and publication figure standards	PPTX
7	Assessment Rubric	100-point rubric with per-criterion breakdown	XLSX
8	Student Reviewer Form	One-page blind review form (see Section 15)	PDF
9	Author Declaration Form	Originality and publication terms agreement	DOCX
10	Contribution Statement Template	Per-author contribution breakdown	DOCX
11	Phase 1 Certificate Template	Digitally issuable, lists certified competencies	DOCX
12	Publication Acknowledgment Letter	For accepted authors, includes DOI and issue reference	DOCX
13	Student Tracking Sheet	Columns: student name, grades per deliverable, attendance %	XLSX
14	Suggested Journals List	10+ journals: name, ISSN, indexing, author guidelines link	XLSX

16. Post-Program Follow-Up (3-Month Period)

YCRJ's responsibility to participants does not end at the close of Phase 3. The organizing team commits to a structured **3-month follow-up period** after the inaugural cohort's program ends, covering all papers submitted to external journals or under YCRJ review.

1. **Monitor journal decisions daily.** The operations coordinator checks the submission portal email accounts daily for any decision notifications (acceptance, revision request, or rejection) on papers submitted externally.

2. **Communicate and explain every decision.** When a decision arrives, the coordinator translates and explains it in full to the student group — including what "minor revision," "major revision," and "reject and resubmit" mean in practice, and what a realistic response timeline looks like.

3. **Hold a dedicated revision session per group.** For any paper requiring revisions, the coordinator schedules a **one-hour individual session** with the group to plan the response. The session produces a structured response-to-reviewers table with three columns: *Reviewer comment — Author response — Revision location in paper.*

4. **Support resubmission.** The coordinator reviews the revised paper before resubmission and confirms that all reviewer comments have been addressed. Final resubmission is handled by the group with coordinator oversight.

5. **Assist with publication agreements.** Upon acceptance, the coordinator reads the publication agreement with the student group and clarifies all terms (copyright, open-access licensing, author rights) before any document is signed.

6. **Compile a final program report.** At the close of the follow-up period, the coordinator prepares a structured report covering: total participants enrolled, Phase 1 completion rate, Phase 2 papers submitted, Phase 3 papers published in YCRJ, and papers submitted to external venues with outcomes. This report informs Cohort 2 planning and supports institutional accreditation reporting.

★ Response-to-Reviewers Template Structure

Every group that receives reviewer comments is given the following three-column table to organize their response before resubmission:

Reviewer Comment	Author Response	Revision in Paper
(Paste exact reviewer comment)	(Explain how and why you addressed it)	(Page, section, and line where change appears)

This format mirrors the response conventions used by IEEE, ACM, and Springer, giving students authentic experience of the revision cycle.

17. Conclusion

The Youth Computing Research Journal is not a workshop, a seminar series, or a student club with aspirations. It is a complete, operational research publication system designed specifically for undergraduate computing students — one that produces real papers, runs a real peer review process, and publishes in a real indexed journal.

Every element of YCRJ — from the modular curriculum to the LaTeX typesetting requirement to the blind review process to the operational guidance for coordinators — is intentional. We have designed a program that takes students seriously as junior researchers and gives them the tools, structure, support, and venue to prove it.

We respectfully submit this proposal for consideration and welcome the opportunity to discuss institutional support, faculty partnership, and resource allocation that would enable YCRJ to launch its inaugural cohort.

Youth Computing Research Journal — Initiative Proposal
Submitted April 2026 — Version 2.0, Enhanced Operational Edition